

Applied Natural Language Processing

Final Exam Sample

1. Describe how word embeddings are used in deep learning models. Why are they critical in NLP pipelines? Compare one-hot encoding with word embeddings. How do their properties impact model training?
2. Explain what it means for Word2Vec to capture semantic similarity. Provide an example of a word analogy.
3. Compare GloVe with word2vec in terms of training methodology and output structure.
4. What is the role of pre-trained embeddings in transfer learning for NLP?
5. Explain why dimensionality reduction (e.g., PCA or t-SNE) is often applied to word embeddings.
6. What are contextual embeddings? How do they differ from static embeddings like word2vec?
7. Discuss the intuition behind the skip-gram model in word2vec. Why is it more effective than CBOW in some tasks?
8. Describe subword embeddings (e.g., FastText). How do they help with out-of-vocabulary (OOV) words?
9. How can learned embeddings be visualized and interpreted? What does clustering tell us about semantic relationships?
10. What is the role of negative sampling in training word2vec? How does it make the training process more efficient compared to traditional softmax?
11. Define precision, recall, and F1 score. When is each metric most informative in NLP tasks?
12. How is accuracy misleading in imbalanced classification problems? Provide an NLP example.
13. Explain how the softmax function converts logits into probabilities.
14. What is cross-entropy loss? Why is it well-suited for classification tasks in NLP?
15. Describe how BLEU score evaluates machine translation. What are its limitations?
16. Explain perplexity as an evaluation metric for language models. How is it computed and interpreted?
17. Describe how Mean Reciprocal Rank (MRR) is computed and interpret its values. In what settings is MRR more appropriate than accuracy or F1 score?
18. Explain how ROUGE is used to evaluate summarization models. What are its limitations?
19. Define KL divergence in the context of NLP. How does it differ from traditional distance metrics?

20. Discuss the use of validation sets and early stopping in training NLP models.
21. Explain the idea of teacher forcing during training. Why is it useful, and what are its pitfalls?
22. What is the role of dropout in deep learning models? How does it help prevent overfitting, and what are its limitations?
23. Describe the role of max pooling in convolutional neural networks (CNNs). How does it affect feature maps, and why is it useful?
24. Compare convolutional layers to fully connected layers in terms of parameter sharing and efficiency.
25. Detail the output size computation in CNNs when multiple filters and strides are applied. Walk through an example using a 32×32 image.
26. Describe why deeper CNNs can extract more complex features. Relate this to NLP feature representations.
27. What is the role of padding in convolutional layers? Compare "valid" and "same" padding strategies and their impact on feature maps.
28. What is the function of the forget gate in an LSTM? Explain how it helps manage long-term dependencies.
29. Explain why GRUs may be preferred over LSTMs in certain sequence modeling tasks. Discuss their architectural differences, efficiency, and implications for training.
30. Compare GRUs and LSTMs: architecture, learnability, and performance trade-offs.
31. What is the vanishing gradient problem in RNNs? How does it affect long sequence training?
32. Explain how RNNs process sequences using hidden states. Why is this design suited to NLP?
33. Discuss the use of RNNs and LSTMs in image processing. In what scenarios are they appropriate, and why are they less common than CNNs?
34. What problem does the attention mechanism solve in encoder-decoder models? Explain with an example.
35. How does multi-head attention work in transformers, and why is it effective?
36. Describe the role of positional encoding in transformers. Why is it necessary?
37. Explain how self-attention differs from cross-attention in seq2seq models.
38. What is layer normalization, and why is it used in transformer blocks?
39. Describe the structure and flow of data through a transformer encoder layer.
40. Discuss the role of feedforward networks within each transformer layer. What purpose do they serve?
41. Explain the concept of causal masking in transformer decoders. Why is it critical for autoregressive tasks?
42. What are the challenges of scaling transformers to longer sequences? How have recent architectures addressed this?

43. Compare transformers with RNNs for sequence modeling. What are the main advantages and drawbacks of each?
44. Explain the role of key, query, and value matrices in attention mechanisms. How do they interact to compute attention weights?
45. How do transformers handle long-range dependencies better than RNNs? What are the limitations of transformers in extremely long sequences?
46. What is the role of pre-normalization in transformer architectures? How does it differ from post-normalization, and why is it often preferred?
47. How does BERT differ from traditional language models? Explain masked language modeling.
48. What is the purpose of the [CLS] token in BERT-based classification models?
49. Explain how BERT embeddings can be used for sentence similarity tasks.
50. Describe the difference between sequence labeling and sequence classification tasks in NLP.
51. What is fine-tuning in the context of pre-trained models? How does it differ from feature extraction?
52. What is the difference between masked language modeling (MLM) and next-sentence prediction (NSP)? Why did BERT use both, and how has this changed in later models?
53. Describe the structure of a seq2seq model with attention used in summarization.
54. Explain the principle behind contrastive learning. How does it differ from supervised learning?
55. Discuss how positive and negative pairs are formed in contrastive learning frameworks. Why is their careful selection important?
56. Describe the training objective of CLIP. How does it unify image and text modalities?
57. Why does CLIP use contrastive learning instead of directly regressing image and text embeddings?
58. Explain how CLIP enables zero-shot classification without explicit fine-tuning.
59. Discuss the role of the temperature parameter in CLIP's contrastive loss. How does it affect learning dynamics? What are some of the challenges and limitations of using CLIP in real-world applications?
60. Explain the intuition behind skip connections (residual connections). Why are they critical in deep networks?
61. How do skip connections alleviate the vanishing gradient problem during backpropagation?
62. Compare plain deep networks and ResNets. How does the presence of skip connections change the optimization landscape?
63. Discuss how skip connections affect representational capacity and network depth in practice.
64. What problem settings are Siamese networks particularly well suited for? Give two real-world examples.
65. Describe how Siamese networks handle the concept of "similarity" instead of traditional classification.

66. Compare a contrastive loss versus a triplet loss in the training of Siamese networks. When might one be preferred over the other?
67. Describe the role of metric-based methods (e.g., prototypical networks) in few-shot learning. How do they simplify classification?
68. How do ensemble methods or meta-learning approaches improve few-shot learning performance under extremely low-data scenarios?
69. What distinguishes GPT-style autoregressive models from masked language models like BERT?
70. Explain how GPT generates text using autoregression. What role does causal masking play?
71. What are the strengths and limitations of GPT for zero-shot and few-shot inference?
72. Describe how GPT-like models use tokenization and positional encoding.
73. Explain the concept of prompt engineering in GPT-based models. How can prompts be designed to elicit desired outputs?
74. What is self-supervised learning? How does it differ from supervised and unsupervised learning?
75. Explain the pretraining-finetuning pipeline in modern self-supervised NLP systems.
76. What are n-grams and how are they used to model language statistically?
77. Compare different approaches for computing sentence embeddings.
78. What is part-of-speech (POS) tagging, and why is it fundamental in NLP?
79. Describe rule-based vs machine learning-based approaches to POS tagging. What are their strengths?
80. Few-shot prompting vs fine-tuning: When to prefer prompting only?